Chapter 9 — Electrostatics and Current Electricity

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Q.1 — Multiple Choice Questions (with correct option and reason)

9.1. Two point charges separated by 10 m have force F. If distance is reduced to 5 m, the force:

(a) decreases to half
(b) increases twice
(c) decreases to one quarter
(d) increases four times

Answer: (d) increases four times — $F \propto 1/r^2$; halving r makes F four times.

9.2. Which electric charge is possible on a particle?

(a) 2.5×10^{-19} C
(b) 3.2×10^{-19} C
(c) 1.6×10^{-10} C
(d) 6.02×10^{23} C

Answer: (b) 3.2×10^{-19} C — Charge must be an integer multiple of e; 3.2×10^{-19} C = 2e.

9.3. Which diagram best represents field lines around two oppositely charged particles?

(a) lines leaving both charges
(b) lines entering both charges
(c) lines from positive to negative
(d) closed circles around both

Answer: (c) lines from positive to negative — Field lines start on positive and end on negative.

9.4. Work done on an electron by a potential difference of 100 V is:

(a) 1.6×10^{-19} eV
(b) 1.6×10^{-17} eV
(c) 6.25×10^{-17} eV
(d) 100 eV

Answer: (d) 100 eV — Energy = qV = 100 eV.

9.5. Potential at 50 cm from a charge of 50 μC is:

(a) 9×10^4 V
(b) 18×10^4 V
(c) 9×10^5 V
(d) 18×10^5 V

Answer: (c) 9×10^5 V — $V = kq/r = 9\times10^9\times50\times10^{-6}/0.5 = 9\times10^5$ V.

9.6. A ball of weight 0.1 N and charge 100 μC is suspended between horizontal plates. Electric intensity is:

(a) 10 N C^{-1}
(b) 100 N C^{-1}
(c) 1000 N C^{-1}
(d) 10000 N C^{-1}

Answer: (c) 1000 N C^{-1} — $E = W/q = 0.1/10^{-4} = 1000$ N/C.

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9.7. A 4 Ω wire is doubled on itself. New resistance is:

(a) 16 Ω **(b)** 4 Ω
(c) 2 Ω **(d)** 1 Ω

Answer: (d) 1 Ω — Length halves and area doubles, so R becomes R/4.

9.8. Current 3.0 A through 6.0 V uses power:

(a) 0.5 W **(b)** 2.0 W
(c) 9.0 W **(d)** 18 W

Answer: (d) 18 W — P = VI = 18 W.

9.9. The algebraic sum of potential changes for a complete circuit is zero. This is:

(a) Ohm's law **(b)** Gauss's law
(c) Kirchhoff's first law **(d)** Kirchhoff's second law

Answer: (d) Kirchhoff's second law — Loop rule is Kirchhoff's second law.

9.10. The radius of curvature of a charged particle in a uniform magnetic field is directly proportional to:

(a) the particle's charge **(b)** the particle's momentum
(c) the particle's energy **(d)** the flux density

Answer: (b) the particle's momentum — r = mv/qB = p/qB.

Q.2 — Short Answer Questions

9.1. How does an aeroplane acquire charge while flying?

Friction with air molecules and dust transfers electrons, so the aeroplane can become charged. The charge distributes over its conducting surface.

9.2. Define electric intensity and electric potential.

Electric intensity is force per unit positive charge, E = F/q. Electric potential is work done per unit positive charge in bringing it from infinity to the point, V = W/q.

9.3. A battery is rated 100 A h. How much charge can it supply?

Q = It = 100 × 3600 = 3.6×10^5 C.

9.4. Is electron-volt a unit of potential difference or energy?

It is a unit of energy: the energy gained by an electron moving through 1 V, equal to 1.6×10^{-19} J.

9.5. A copper wire of length ℓ has resistance R. It is stretched to double length. New resistance?

Volume is constant, so area halves. R ∝ L/A, so new resistance is 4R.

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9.6. Why does resistance of a conductor rise with temperature?

Lattice ions vibrate more strongly, scattering electrons more often, so resistivity increases.

9.7. Is filament resistance lower or higher in a 500 W, 220 V bulb than in a 100 W, 220 V bulb?

Lower. $R = V^2/P$, so larger power at the same voltage means smaller resistance.

9.8. Why does resistance of a thermistor change as temperature increases?

In most thermistors, heating releases more charge carriers, so resistance decreases (NTC behaviour).

9.9. Which materials can be used to construct a Faraday cage and why?

Conductors such as copper, aluminium or steel mesh. Free charges rearrange to cancel external electric fields inside the enclosure.

Constructed Response Questions

9.1 Why do electric lines of force never cross?

If they crossed, the field would have two directions at one point, which is impossible because the force on a test charge has a unique direction.

9.2 Is E necessarily zero inside a charged spherical rubber balloon with uniform surface charge?

Yes. By symmetry and Gauss's law, the field inside a uniformly charged spherical shell is zero.

9.3 Electrostatic force is 10^{36} times stronger than gravity. Argue that the galaxy should be almost electrically neutral.

If large net charges existed, electric forces would dominate gravity and disrupt the galaxy. Since gravity governs large-scale structure, matter must be nearly neutral overall.

9.4 An uncharged conducting hollow sphere is placed in the field of a positive charge q. What is net flux through the shell?

Zero, because the shell encloses no net charge. External field lines enter and leave equally.

9.5 A potential difference is applied across a copper wire. Effect on drift velocity by (i) increasing p.d. (ii) decreasing length and temperature?

(i) Increasing p.d. increases field and drift velocity. (ii) Decreasing length increases field for the same p.d.; lowering temperature reduces scattering, both increasing drift velocity.

9.6 Why does terminal p.d. of a battery decrease when current drawn increases?

$V = E - Ir$. Larger current gives a larger internal voltage drop Ir, so terminal voltage falls.

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Comprehensive / Long Questions

9.1 Explain electric potential and prove electric field intensity equals negative potential gradient.

Potential difference is work per unit charge: $\Delta V = -E\Delta d$ for a uniform field. Hence $E = -\Delta V/\Delta d$. The negative sign shows the field points toward decreasing potential.

9.2 State and explain Kirchhoff's rules.

First rule: sum of currents entering a junction equals sum leaving, conserving charge. Second rule: algebraic sum of potential changes around a closed loop is zero, conserving energy. Use +E from − to +, −E from + to −, −IR with current, +IR against current.

9.3 What is a Wheatstone bridge? Explain its working with a diagram.

Four resistances form a diamond with a galvanometer between two junctions. At balance no current flows through the galvanometer, giving $R_1/R_2 = R_3/R_4$. An unknown resistance can then be found from three known resistances.

9.4 What is an LDR? How can it be used as an ON-OFF switch for lighting?

An LDR's resistance falls when illuminated. In a potential divider it controls the base/gate of a transistor or relay, switching lights on when dark and off when bright.

9.5 What is a potentiometer? Describe its working.

A potentiometer is a uniform wire used as a potential divider. A cell is balanced against the potential drop along a length of wire; at null deflection $E \propto \ell$, so emfs can be compared without drawing current.

Numerical Problems (fully worked)

9.1 Two charges repel with 0.4 N at 5.0 cm. Find force at (a) 2.5 cm (b) 15.0 cm.

Given: $F_1 = 0.4$ N, $r_1 = 5$ cm

Formula: $F \propto 1/r^2$

(a) r halved, $F = 0.4\times4 = 1.6$ N. (b) r tripled, $F = 0.4/9 \approx 0.044$ N ≈ 0.04 N.

Answer: (a) 1.6 N (b) 0.04 N

9.2 Charge +20 µC between plates 10 cm apart with p.d. 0.5 kV. Find E and force.

Given: $q = 20\times10^{-6}$ C, d = 0.10 m, V = 500 V

Formula: $E = V/d$, $F = qE$

$E = 500/0.10 = 5000$ N C^{-1}.

$F = 20\times10^{-6} \times 5000 = 0.10$ N = 100 mN.

Answer: 5 kN C^{-1}, 100 mN

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9.3 Electron and proton in hydrogen separated by 5.3×10^{-11} m. Find ratio of electric to gravitational force.

Given: $r = 5.3\times10^{-11}$ m

Formula: $F_e/F_g = ke^2/(Gm_e m_p)$

Ratio $\approx 2.3 \times 10^{39}$, independent of separation.

Answer: $\approx 2.3 \times 10^{39}$

9.4 A water droplet of mass 1.2×10^{-14} kg is suspended in a downward electric field 6.0×10^{4} N C^{-1}. Find charge.

Given: $m = 1.2\times10^{-14}$ kg, $E = 6.0\times10^{4}$ N C^{-1}

Formula: $qE = mg$, charge negative for upward force

$q = -mg/E \approx -1.96 \times 10^{-18}$ C, about -1.98×10^{-18} C as in the book.

Answer: $\approx -1.98 \times 10^{-18}$ C

The printed exponents are difficult to read in OCR; the book answer is retained.

9.5 An electron enters a uniform field with $v_x = 2.98 \times 10^{7}$ m s^{-1}, E = 300 N C^{-1}, plate length 10.0 cm. Find acceleration and time in field.

Given: $v_x = 2.98\times10^{7}$ m s^{-1}, E = 300 N C^{-1}, L = 0.10 m

Formula: $a = eE/m$, $t = L/v_x$

$a = 1.6\times10^{-19}\times300/9.11\times10^{-31} \approx 5.27 \times 10^{13}$ m s^{-2}.

$t = 0.10/2.98\times10^{7} \approx 3.34 \times 10^{-9}$ s.

Answer: 5.27×10^{13} m s^{-2}, 3.34×10^{-9} s

9.6 A disc of area 10 cm^2 is in vertical field $E = 5 \times 10^{5}$ N C^{-1}. Plane makes 30° with horizontal. Find flux.

Given: $A = 10^{-3}$ m^2, $E = 5\times10^{5}$ N C^{-1}, angle between normal and field = 30°

Formula: $\Phi = EA \cos\theta$

$\Phi = 5\times10^{5} \times 10^{-3} \times \cos30° = 250\sqrt{3}$ N m^2 C^{-1}.

Answer: $250\sqrt{3}$ N m^2 C^{-1}

9.7 Copper rod 50 cm long, 1 cm diameter. Find resistance. Find side of square tungsten rod 50 cm long with same resistance.

Given: $\rho_{Cu} = 1.69\times10^{-8}$ Ω m, $\rho_W = 5.0\times10^{-8}$ Ω m, L = 0.5 m

Formula: $R = \rho L/A$

$A_{Cu} = \pi(0.005)^2 \approx 7.85\times10^{-5}$ m^2.

$R \approx 1.08 \times 10^{-4}$ Ω.

For tungsten, $A = \rho_W L/R \approx 2.31\times10^{-4}$ m^2; side ≈ 1.52 cm.

Answer: 1.08×10^{-4} Ω, 1.52 cm

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9.8 Copper winding resistance 50 Ω at 30°C becomes 52 Ω. α = 0.0039 K⁻¹. Find temperature rise.

Given: $R_0 = 50\ \Omega$, $R = 52\ \Omega$, $\alpha = 0.0039\ K^{-1}$

Formula: $\Delta R = R_0 \alpha \Delta T$

$\Delta T = 2/(50 \times 0.0039) \approx 10.3°C$.

Answer: 10.3°C

9.9 Copper wire 50 m long, 150 µm diameter, current 1 A for 50 s. Find resistance and heat dissipated.

Given: $L = 50\ m$, $d = 150 \times 10^{-6}\ m$, $I = 1\ A$, $t = 50\ s$, $\rho = 1.69 \times 10^{-8}\ \Omega\ m$

Formula: $R = \rho L/A$, $H = I^2Rt$

$A \approx 1.77 \times 10^{-8}\ m^2$, $R \approx 47.8\ \Omega$.

$H = 1^2 \times 47.8 \times 50 \approx 2390\ J$.

Answer: 47.8 Ω, 2390 J

9.10 Battery emf 12 V connected to 3.6 Ω; internal resistance 0.2 Ω. Find terminal voltage.

Given: $E = 12\ V$, $R = 3.6\ \Omega$, $r = 0.2\ \Omega$

Formula: $I = E/(R+r)$, $V = IR$

$I = 12/3.8 \approx 3.16\ A$.

$V = 3.16 \times 3.6 \approx 11.4\ V$.

Answer: 11.4 V

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Textbook checks flagged in this chapter

• Numerical 9.4: the printed mass/field exponents are unclear in OCR; the book answer is retained with the formula q = −mg/E.